Visual Biomechanics and Exercise Decisions Solutions

HN-FE02 | Instructor key | 100 points | 120 minutes

Award the stated subpart points for correct calculations with units and requested reasoning. Accept defensible alternatives in design questions. Numeric tolerance 0.05 unless an exact count is required. Keep this key separate from student sources.

Question 1 Read the geometry

HN-FE02-Q01 | 10 points

a) [4 points] Force 140 N; radius 0.50 m; perpendicular distance 0.30 m.

b) [3 points] 140×.30=42 N·m.

c) [3 points] 140×.10=14 N·m; one-third of the original component. Internal muscle forces require additional anatomy/forces/dynamics.

Course modules 4.

Question 2 Two joint anatomy

HN-FE02-Q02 | 10 points

a) [4 points] Hip flexion tends to lengthen biarticular hamstrings across hip and shorten rectus femoris across hip.

b) [3 points] Biceps femoris short head.

c) [3 points] Curl is knee flexion; hip thrust is hip extension. A shared region tag does not preserve joint action.

Course modules 4.

Question 3 Compare profiles from a visual

HN-FE02-Q03 | 10 points

a) [4 points] A 20, 12, 4 N·m; B 8, 24, 16 N·m at 80 N.

b) [3 points] A at start; B at middle and end.

c) [3 points] No; target/joint configuration and longitudinal outcome data are not supplied. Three samples are not a complete continuous curve.

Course modules 4.

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Question 4 Press and pull observations

HN-FE02-Q04 | 10 points

a) [4 points] Overhead elevation permits scapular upward rotation; selected bench setup does not transfer unchanged.

b) [3 points] No angle guarantees absence of injury; reference, setup and individual context matter.

c) [3 points] Retraction moves scapula toward spine; horizontal abduction moves upper arm relative to torso. Grip label alone does not establish future regional growth.

Course modules 4.

Question 5 Nutrition scenario

HN-FE02-Q05 | 15 points

a) [5 points] MSJ=10(72)+6.25(170)-5(30)-161=1471.5 kcal/day. TDEE=1471.5×1.5=2207.25; planned intake=2207.25×1.05=2317.6125 kcal/day.

b) [5 points] Protein=1.8×72=129.6 g; fat=0.8×72=57.6 g. Their energy=4(129.6)+9(57.6)=1036.8 kcal. C=(2317.6125-1036.8)/4=320.203125 g.

c) [5 points] Branch, multiplier, macro factors and intake adjustment are assumptions; no measured expenditure or local growth response is supplied.

Course modules 1, 2, 5.

Question 6 Volume and frequency

HN-FE02-Q06 | 15 points

a) [5 points] D 3, I 6; counts 3, 6, 9.

b) [5 points] Direct 2 days(Mon, Fri); contributing 3(Mon, Wed, Fri).

c) [5 points] Several exercise rows on one day are one exposure day; compound sets may contribute to multiple targets.

Course modules 3.

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Question 7 Progression and a conditional block

HN-FE02-Q07 | 15 points

a) [5 points] Propose 42.5 kg total; 6.25% increase.

b) [5 points] 7, 8, 9, 9, 10, 5; total 48 proposed direct sets.

c) [5 points] No; rebaseline the changed task. Review comparable performance, recovery, adherence or execution before adding.

Course modules 3, 4, 5.

Question 8 Substitution and scheduling

HN-FE02-Q08 | 15 points

a) [5 points] Both elbow extension; overhead dumbbell retains broad overhead shoulder position while pressdown changes it. Force/profile/support require reassessment; no clinical clearance follows.

b) [5 points] 20-minute overlap; 7-hour sleep opportunity. Move cycling after the resistance session or another feasible slot; time alone does not certify recovery.

c) [5 points] Overgeneralization: the discussed primary comparison favored standing for gastrocnemius while soleus changes were similar; retain muscle-specific scope.

Course modules 4, 5.